

# Master's-Level Interview Preparation Guide: Fluid Mechanics & Heat Transfer for Fusion, Materials, and Space/Aerospace Roles

## TL;DR

- This guide organizes graduate-level thermal-fluids into concept reviews + governing equations + worked interview questions across three industry lenses: **fusion** (MHD liquid-metal blankets,  $\sim 10 \text{ MW/m}^2$  divertor heat flux, cryogenic magnets), **materials** (thermal shock/fatigue, conductivity/diffusivity, high-T behavior), and **space** (radiation-dominated vacuum thermal control, re-entry aerothermal heating, microgravity boiling).
  - The single highest-leverage differentiator by sector: fusion candidates must master **MHD pressure drop** ( $\propto B^2$  for conducting walls, the reason for insulating inserts); space candidates must master **radiation exchange** ( $q \propto \sigma T^4$ , no convection in vacuum); materials candidates must master **transient conduction, Biot number, and thermal-stress merit indices**.
  - Approach open-ended questions with **order-of-magnitude reasoning**: identify the dominant transport mode via dimensionless groups (Re, Pr, Bi, Ra, Ha, N), write the governing balance, non-dimensionalize, and sanity-check against limiting cases before computing.
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## Key Findings (How to Use This Guide)

1. **Interviews test reasoning, not memorization.** You will be judged on whether you can identify the controlling physics (which dimensionless number dominates), write the correct governing equation, and estimate a number.
  2. **Sector signal words:** "blanket / breeder / Hartmann / PbLi / FLiBe / divertor"  $\rightarrow$  fusion; "MLI / radiator / view factor / re-entry / ablation / cryo-propellant"  $\rightarrow$  space; "thermal shock / diffusivity / CTE mismatch / creep"  $\rightarrow$  materials.
  3. **The universal toolkit** (continuity, Navier-Stokes, energy equation, boundary layers, dimensionless groups) underlies all three; industry context changes only which terms dominate.
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## PART I — FLUID MECHANICS

### 1. Conservation Laws & Control-Volume Analysis

**Concepts.** All of continuum fluid mechanics rests on conservation of mass, momentum, and energy applied either to a control volume (integral form) or a fluid element (differential form).

#### Governing equations.

- Continuity (incompressible):  $\nabla \cdot \mathbf{u} = 0$ . Compressible:  $\partial \rho / \partial t + \nabla \cdot (\rho \mathbf{u}) = 0$ .

- Navier-Stokes (incompressible, constant  $\mu$ ):  $\rho(\partial \mathbf{u}/\partial t + \mathbf{u} \cdot \nabla \mathbf{u}) = -\nabla p + \mu \nabla^2 \mathbf{u} + \rho \mathbf{g}$ .
- Reynolds Transport Theorem links system to control volume:  $dB_{\text{sys}}/dt = \partial/\partial t \int_{\text{CV}} \rho b \, dV + \int_{\text{CS}} \rho b (\mathbf{u} \cdot \mathbf{n}) dA$ .
- Energy (thermal):  $\rho c_p (DT/Dt) = \nabla \cdot (k \nabla T) + \Phi$  (viscous dissipation) +  $\dot{q}$ .

**Interview Q (conceptual):** *"Physically interpret each term in Navier-Stokes."* **Answer:** LHS is inertia (unsteady + convective acceleration); RHS terms are pressure gradient (drives flow), viscous diffusion of momentum, and body force. The ratio of convective inertia to viscous term is the Reynolds number.

**Interview Q (quantitative):** *"Water flows through a reducing bend; how do you find the force on the bolts?"* **Answer outline:** Apply integral momentum to a CV around the bend:  $\Sigma F = \dot{m}(\mathbf{u}_{\text{out}} - \mathbf{u}_{\text{in}}) + \text{pressure-area terms}$ . Include gauge pressures at inlet/outlet, resolve into x and y, and the reaction force on the fluid is equal-and-opposite to what the bolts carry. Emphasize choosing the CV and sign conventions first.

## 2. Dimensional Analysis & Dimensionless Groups

**Concepts.** Buckingham- $\Pi$  theorem: a problem with  $n$  variables and  $k$  fundamental dimensions reduces to  $(n-k)$  dimensionless groups. Master the physical *ratio* each group represents — interviewers love "what does X number physically mean?"

**Reference table of key groups:**

| Group                | Definition                                    | Physical meaning                                  | Where it dominates     |
|----------------------|-----------------------------------------------|---------------------------------------------------|------------------------|
| Reynolds Re          | $\rho UL/\mu$                                 | inertia / viscous                                 | transition, all flows  |
| Prandtl Pr           | $\nu/\alpha = \mu c_p/k$                      | momentum diff / thermal diff                      | BL thickness ratio     |
| Nusselt Nu           | $hL/k$                                        | convective / conductive HT at wall                | all convection         |
| Péclet Pe            | $RePr = UL/\alpha$                            | advective / diffusive heat                        | thermal entry          |
| Grashof Gr           | $g\beta\Delta T L^3/\nu^2$                    | buoyancy / viscous                                | natural convection     |
| Rayleigh Ra          | $Gr\cdot Pr = g\beta\Delta T L^3/(\nu\alpha)$ | buoyancy vs diffusion                             | onset of convection    |
| Mach Ma              | $U/a$                                         | flow speed / sound speed                          | compressibility        |
| Weber We             | $\rho U^2 L/\sigma$                           | inertia / surface tension                         | atomization, sprays    |
| Froude Fr            | $U/\sqrt{gL}$                                 | inertia / gravity                                 | free surface           |
| Stanton St           | $h/(\rho c_p U) = Nu/(RePr)$                  | HT to wall / thermal capacity of flow             | HX, boundary layers    |
| Biot Bi              | $hL_c/k_{solid}$                              | internal / surface resistance                     | transient conduction   |
| Fourier Fo           | $\alpha t/L^2$                                | dimensionless time                                | transient conduction   |
| Hartmann Ha          | $BL\sqrt{(\sigma/\mu)}$                       | $(EM/viscous)^{1/2}$ ( $Ha^2 = Lorentz/viscous$ ) | MHD wall layers        |
| Stuart N             | $Ha^2/Re = \sigma B^2 L/(\rho U)$             | Lorentz / inertia                                 | MHD core flow          |
| Magnetic Reynolds Rm | $\mu_0 \sigma UL$                             | advection / diffusion of B                        | dynamo vs quasi-static |

**Interview Q (gotcha):** "Prandtl number of liquid metals is  $\sim 0.01$ . What does that tell you?"

**Answer:** Thermal diffusivity  $\gg$  momentum diffusivity, so the thermal boundary layer is much thicker than the velocity boundary layer. Heat conducts away from the wall far faster than momentum diffuses. Consequence: standard correlations (Dittus-Boelter) developed for  $Pr \sim 1$  fail; you use liquid-metal correlations (e.g.,  $Nu = 4.82 + 0.0185 Pe^{0.827}$  for constant  $q''$ , Skupinski). Highly relevant to fusion liquid-metal coolants and to sodium-cooled systems.

### 3. Laminar/Turbulent Flow, Boundary Layers & Transition

**Concepts.** Prandtl's boundary-layer concept: viscous effects confined to a thin layer near walls; outside it flow is ~inviscid.

#### Governing/results.

- Blasius laminar flat plate:  $\delta/x = 5.0/\sqrt{\text{Re}_x}$ ;  $\delta \propto \sqrt{x}$  (grows parabolically); (Fiveable) local skin friction  $c_f = 0.664/\sqrt{\text{Re}_x}$ .
- Displacement thickness  $\delta^*$  and momentum thickness  $\theta$  (von Kármán integral relation ties  $\theta$  to wall shear/drag). (Fiveable)
- Transition on a flat plate: commonly  $\text{Re}_{x,\text{crit}} \approx 5 \times 10^5$  (engineering value); (Utexas) linear instability onset  $\text{Re} \sim O(10^5)$  (arxiv) via Tollmien-Schlichting waves; bypass transition can occur earlier with free-stream turbulence. (Fiveable)
- Turbulent BL:  $\delta/x \approx 0.16/\text{Re}_x^{(1/7)}$ ; fuller profile, higher skin friction but more resistant to separation. (Fiveable)

**Interview Q:** "Why does a golf ball have dimples?" **Answer:** Dimples trip the boundary layer to turbulent, which has more near-wall momentum and resists separation, moving the separation point rearward  $\rightarrow$  smaller wake  $\rightarrow$  lower pressure (form) drag. Net drag drops despite higher skin friction. This is the laminar-vs-turbulent separation trade-off.

### 4. Viscous Internal (Pipe) Flow

**Concepts.** Laminar ( $\text{Re} < 2300$ ), transitional, turbulent ( $\text{Re} > 4000$ ) pipe flow. Fully developed vs entrance region.

#### Governing/results.

- Darcy-Weisbach:  $\Delta p = f(L/D)(\rho U^2/2)$ ; head loss  $h_f = f(L/D)(U^2/2g)$ .
- Laminar:  $f = 64/\text{Re}$  (Hagen-Poiseuille);  $\text{Nu} = 4.36$  (constant  $q''$ ),  $3.66$  (constant  $T_w$ ).
- Turbulent friction factor from Colebrook equation / Moody chart (function of  $\text{Re}$  and relative roughness  $\epsilon/D$ ).
- Minor losses via loss coefficients  $K$ .

**Interview Q (quantitative):** "Estimate pressure drop for water in a 5 cm pipe, 100 m long, at 2 m/s." **Answer outline:**  $\text{Re} = UD/\nu \approx (2)(0.05)/(1\text{e-}6) = 1 \times 10^5 \rightarrow$  turbulent. Smooth pipe  $f \approx 0.018$  (Moody).  $\Delta p = f(L/D)(\rho U^2/2) = 0.018(100/0.05)(1000 \cdot 4/2) = 0.018 \cdot 2000 \cdot 2000 \approx 72 \text{ kPa}$ . State assumptions (smooth, isothermal, fully developed).

### 5. Compressible Flow (Aerospace emphasis)

**Concepts.** When  $\text{Ma} \gtrsim 0.3$ , density changes matter. Isentropic flow, normal/oblique shocks, nozzle flow.

#### Governing/results.

- Speed of sound  $a = \sqrt{\gamma RT}$ ; stagnation relations  $T_0/T = 1 + (\gamma-1)/2 \cdot \text{Ma}^2$ .

- Isentropic  $p_0/p = (1 + (\gamma - 1)/2 Ma^2)^{\gamma/(\gamma - 1)}$ .
- Area-Mach relation: converging-diverging nozzle chokes at throat ( $Ma=1$ ,  $A=A^*$ ); [\(Purdue University\)](#) supersonic flow only in diverging section. [\(Purdue University\)](#)
- Normal shock: downstream subsonic, stagnation pressure drops (entropy rise), stagnation temperature constant, mass flow conserved. [\(Psesht\)](#)

**Interview Q (gotcha):** "In a choked converging nozzle, what happens to mass flow if you lower back pressure further?" **Answer:** Nothing — once choked ( $Ma=1$  at throat), mass flow is fixed by upstream stagnation conditions; downstream disturbances can't propagate upstream past the sonic throat. [\(Psesht\)](#) To increase  $\dot{m}$  you must raise  $p_0$  or lower  $T_0$  (or enlarge throat).

**Interview Q:** "CD nozzle with a normal shock in the diverging section — walk through the flow." **Answer:** Subsonic in converging section  $\rightarrow$  sonic at throat  $\rightarrow$  supersonic isentropic acceleration  $\rightarrow$  non-isentropic normal shock [\(arxiv\)](#) (jumps to subsonic,  $p$  rises,  $p_0$  drops)  $\rightarrow$  subsonic diffusion to exit. Shock location set by back pressure. Lowering back pressure pushes shock downstream toward exit. [\(Compressible Flow\)](#)

## 6. Magnetohydrodynamics (MHD) — Fusion Critical

**Concepts.** Flow of electrically conducting fluid (liquid metals Li, PbLi; molten salts partly) in a magnetic field. Motion across  $\mathbf{B}$  induces currents  $\mathbf{J} = \sigma(\mathbf{E} + \mathbf{u} \times \mathbf{B})$ ; these currents interact with  $\mathbf{B}$  to produce a retarding Lorentz force  $\mathbf{J} \times \mathbf{B}$ . Consequences for fusion blankets: laminarization of turbulence, flattened core velocity profiles, thin wall boundary layers, and enormous pressure drops.

**Governing equations (quasi-static, low  $Rm$  — appropriate for blankets):**

- Momentum with Lorentz force:  $\rho(D\mathbf{u}/Dt) = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{J} \times \mathbf{B}$ .
- Ohm's law:  $\mathbf{J} = \sigma(-\nabla \phi + \mathbf{u} \times \mathbf{B})$ ; charge conservation  $\nabla \cdot \mathbf{J} = 0 \rightarrow \nabla^2 \phi = \nabla \cdot (\mathbf{u} \times \mathbf{B})$ .
- Key numbers: Hartmann  $Ha = BL\sqrt{\sigma/\mu}$  ( $Ha^2 = \text{Lorentz}/\text{viscous}$ ); Stuart/interaction parameter  $N = Ha^2/Re = \sigma B^2 L / (\rho U)$  [\(Wikipedia\)](#) (Lorentz/inertia); magnetic Reynolds  $Rm = \mu_0 \sigma UL$  [\(arxiv\)](#) (blankets have  $Rm \ll 1$ , so imposed field is undisturbed  $\rightarrow$  quasi-static approximation).

**Hartmann flow (canonical problem).** Conducting fluid between parallel plates, transverse  $B$ . Solution: flat "slug"-like core with thin **Hartmann layers** at walls perpendicular to  $B$ , thickness  $\delta_H \approx L/Ha$ . In ducts, **Shercliff (side) layers** on walls parallel to  $B$  scale as  $\delta_S \approx L/\sqrt{Ha}$ . The Hartmann layer transitions to turbulence at a critical Reynolds number based on the laminar Hartmann-layer thickness of approximately 380 ( $\pm 10\%$ ), a result Moresco & Alboussière (2004, *J. Fluid Mech.* 504:167-181) found to hold for both transition and laminarization across a wide range of magnetic field intensity in mercury with  $B$  up to 13 T.

**MHD pressure drop (the headline result).**

- General form:  $\nabla p = k \cdot \sigma \cdot u \cdot B^2$ , where  $k$  is a dimensionless pressure-gradient coefficient.

- **Conducting walls** (Chang & Lundgren 1961):  $k = c/(1+c)$ , where the **wall conductance ratio**  $c = (t_w \cdot \sigma_w)/(a \cdot \sigma)$  [wall thickness  $\times$  wall conductivity / (channel half-width  $\times$  fluid conductivity)]. So  $\Delta p \propto B^2$  — catastrophic at fusion fields.
- **Insulating walls** (Shercliff 1962):  $k = 3\pi/(8 \cdot Ha)$ , so  $\Delta p \propto B$  (much smaller). This is the entire motivation for **flow-channel inserts (FCIs)** made of SiC or alumina to electrically decouple the flow from load-bearing steel walls.
- The MHD pressure drop can exceed the ordinary hydrodynamic drop by factor  $\sim kHa^2$ , i.e. by 5–7 orders of magnitude in fusion applications (Mistrangelo & Bühler, EUROfusion/KIT).

**Worked example (fusion, quantitative).** PbLi at  $\sim 500^\circ\text{C}$ :  $\sigma \approx 0.77 \times 10^6 \text{ S/m}$ ,  $\rho \approx 9300 \text{ kg/m}^3$ ,  $\mu \approx 1.15 \times 10^{-3} \text{ Pa}\cdot\text{s}$  (Schulz/ORNL correlations; standard database Mas de les Valls et al., *J. Nucl. Mater.* 376 (2008) 353–357). Take  $u = 0.1 \text{ m/s}$  (DCLL poloidal duct velocity, Abdou et al., *Fusion Eng. Des.* 100 (2015) 2–43), channel half-width  $a = 0.1 \text{ m}$  ( $\sim 20 \text{ cm}$  duct), bare conducting steel wall giving  $c \approx 0.08 \rightarrow c/(1+c) \approx 0.08$ .

- At  $B = 4 \text{ T}$ :  $dp/dx \approx 0.08 \cdot (0.77 \times 10^6) \cdot (0.1) \cdot (16) \approx 1.0 \times 10^5 \text{ Pa/m} \rightarrow \sim 1 \text{ MPa}$  over a 10-m poloidal duct.
- At  $B = 10 \text{ T}$ :  $dp/dx \approx 6 \times 10^5 \text{ Pa/m} \rightarrow$  multi-MPa per metre (prohibitive) — hence insulating inserts are mandatory. Reported DCLL blanket totals:  $\sim 0.4 \text{ MPa}$  (outboard) and  $1.17 \text{ MPa}$  (inboard) for US DEMO; design limit historically kept  $\lesssim 2 \text{ MPa}$  (Smolentsev, *Fluids* 6(3):110, 2021).

**Interview Q (fusion gotcha):** "Why is the MHD pressure drop so much worse with metal walls than insulating walls, and what's the scaling with  $B$ ?" **Answer:** With conducting walls, induced currents have a low-resistance return path through the wall, so large currents flow  $\rightarrow$  large  $\mathbf{J} \times \mathbf{B}$  braking force  $\rightarrow \Delta p \propto B^2$  (via  $k = c/(1+c)$ ). Insulating walls force currents to close within the thin fluid layers, drastically reducing current magnitude  $\rightarrow \Delta p \propto B$  ( $k = 3\pi/8Ha$ ). At  $B \sim 4\text{--}10 \text{ T}$  the  $B^2$  scaling is the dominant blanket-feasibility issue; FCIs are used to make  $c \rightarrow 0$ .

## 7. Multiphase / Two-Phase Flow

**Concepts.** Flow regimes (bubbly, slug, annular, mist in vertical pipes; stratified/wavy in horizontal). Void fraction  $\alpha$ , quality  $x$ , slip ratio. Homogeneous vs separated (drift-flux) models. Pressure drop = frictional + accelerational + gravitational components.

**Interview Q:** "Why does two-phase flow pressure drop exceed single-phase?" **Answer:** Higher velocities from vapor generation (acceleration term), increased interfacial friction, and phase distribution effects; commonly captured via two-phase multiplier  $\phi^2_{LO}$  applied to the liquid-only friction gradient (Lockhart–Martinelli / Martinelli parameter  $X$ ).

## 8. Turbulence Modeling Overview

**Concepts.** Turbulence is 3-D, chaotic, multiscale (energy cascade from large eddies to Kolmogorov dissipation scale). Direct Numerical Simulation (DNS) resolves all scales —

prohibitively expensive at high Re.

### Hierarchy:

- **RANS** (Reynolds-Averaged Navier-Stokes): solve time-averaged equations; model all turbulence via a closure. Cheapest.
  - **k- $\epsilon$** : robust for free-shear/internal flows, ([arxiv](#)) good far-field; poor near walls & adverse pressure gradients; needs wall functions.
  - **k- $\omega$** : better in the viscous sublayer and adverse pressure gradients; sensitive to free-stream ([Engineering.com](#))  $\omega$ .
  - **k- $\omega$  SST (Menter)**: hybrid — k- $\omega$  near walls, k- $\epsilon$  in free stream via blending. ([arxiv](#)) Industry workhorse for aerodynamics and separated flows.
  - **Spalart-Allmaras**: one-equation, cheap, good for attached aero flows. ([Engineering.com](#))
- **LES**: filters equations, resolves large energy-containing eddies, models only sub-grid scales. ([ResearchGate](#)) Much costlier than RANS, cheaper than DNS. Not a "turbulence model" but a different formulation. ([ResearchGate](#))
- **Hybrid (DES/SBES)**: RANS near walls, LES away.

**Interview Q:** "When would you pick LES over RANS?" **Answer:** When you need unsteady large-scale structures — separated flows, combustion instabilities, acoustics, bluff-body wakes, or when RANS closures are known to fail. For steady attached flows and quick design iteration, RANS (k- $\omega$  SST) suffices. In fusion MHD, a strong field suppresses/laminarizes turbulence (high Stuart number  $\rightarrow$  quasi-2D), ([ScienceDirect](#)) which changes modeling needs (special low-Rm MHD RANS or quasi-2D models).

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## PART II — HEAT TRANSFER

### 9. Conduction

**Concepts.** Fourier's law  $q'' = -k \nabla T$ . Heat equation, steady & transient, thermal resistance networks (analogy to circuits), fins, lumped capacitance.

#### Governing/results.

- Heat equation:  $\partial T / \partial t = \alpha \nabla^2 T + \dot{q} / (\rho c_p)$ ,  $\alpha = k / (\rho c_p)$ .
- 1-D plane wall resistance  $R = L / (kA)$ ; convective  $R = 1 / (hA)$ ; radial (cylinder)  $R = \ln(r_2/r_1) / (2\pi kL)$ . Series/parallel networks.
- **Fins:**  $m = \sqrt{hP/kA_c}$ ; efficiency  $\eta_f = \tanh(mL) / (mL)$  for adiabatic-tip. Fins help when Bi is small (conduction along fin  $\gg$  convection off it).
- **Lumped capacitance** (valid when  $Bi = hL_c/k < 0.1$ ):  $T(t) - T_\infty = (T_i - T_\infty)e^{-(t/\tau)}$ ,  $\tau = \rho V c_p / (hA_s)$ .
- Transient distributed: use Heisler charts / one-term series in  $Fo = \alpha t / L^2$ .

**Interview Q (gotcha):** "When is the lumped-capacitance approximation valid, and what's the physical meaning of Biot number?" **Answer:** Valid when  $Bi = hL_c/k_{\text{solid}} < 0.1$ , i.e. internal conduction resistance  $\ll$  surface convection resistance, so the body is nearly isothermal internally. High- $k$ , small bodies with modest  $h$  qualify. Contrast Biot (uses  $k$  of the solid) with Nusselt (uses  $k$  of the fluid) — a classic trap.

**Interview Q (materials):** "Two rods, same  $k$  but different  $\rho c_p$  — which heats up faster transiently?" **Answer:** The one with lower  $\rho c_p$  (higher thermal diffusivity  $\alpha = k/\rho c_p$ ) responds faster; steady-state gradient is identical (set by  $k$ ), but transient response is governed by  $\alpha$ . This distinction between conductivity and diffusivity is a favorite materials question.

## 10. Convection (Forced & Natural)

**Concepts.** Newton's law  $q'' = h(T_s - T_\infty)$ .  $h$  obtained from Nu correlations. Distinguish forced (Re, Pr drive) vs natural (Ra drives), internal vs external.

**Key correlations (know assumptions/validity):**

- **Dittus-Boelter** (turbulent internal, smooth tube):  $Nu = 0.023 Re^{0.8} Pr^n$ ,  $n=0.4$  heating /  $0.3$  cooling; (DSPE) valid  $Re > 10,000$ ,  $0.7 < Pr < 160$ ,  $L/D > 10$ . (ChemEnggCalc)
- **Sieder-Tate:**  $Nu = 0.027 Re^{0.8} Pr^{(1/3)} (\mu/\mu_w)^{0.14}$  — corrects for temperature-dependent viscosity (large  $\Delta T$ ). (ChemEnggCalc)
- **Gnielinski:** most accurate turbulent tube correlation, valid through transition (uses friction factor).
- **Churchill-Bernstein:** external cross-flow over a cylinder (all  $Re$ - $Pr > 0.2$ ).
- Laminar fully developed internal:  $Nu = 4.36$  (const  $q''$ ),  $3.66$  (const  $T_w$ ) — constants, independent of  $Re$ ,  $Pr$ .
- Natural convection vertical plate:  $Nu = f(Ra)$ ; e.g. Churchill-Chu.

**Interview Q (gotcha):** "You apply Dittus-Boelter and get a strange  $h$  for laminar flow. What went wrong?" **Answer:** Dittus-Boelter is only valid for turbulent flow ( $Re > 10,000$ ).

(ScienceDirect) For laminar fully-developed flow,  $Nu$  is a fixed constant (4.36 or 3.66) independent of  $Re$  and  $Pr$  — do not use a turbulent correlation. Also check that properties are evaluated at the correct (bulk mean) temperature.

**Interview Q (space):** "Can you use natural convection correlations to cool a box on a satellite?" **Answer:** No — natural convection requires gravity-driven buoyancy. In microgravity (and in vacuum, where there's no fluid) there's no buoyant convection; on orbit, heat rejection is essentially radiation-only. This is *the* space thermal reality check.

## 11. Radiation — Space Critical

**Concepts.** Thermal radiation needs no medium — the *only* steady heat-rejection path in vacuum. Blackbody, graybody, emissivity  $\epsilon$ , view factors  $F_{ij}$ , radiosity networks.

**Governing/results.**

- Stefan-Boltzmann:  $E_b = \sigma T^4$  ( $\sigma = 5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$ ). (NovaSolver Project) Graybody emissive power =  $\epsilon \sigma T^4$ .
- Net exchange (two black surfaces):  $q = A_1 F_{12} \sigma (T_1^4 - T_2^4)$ . (arxiv)
- Graybody radiosity:  $J_i = \epsilon_i \sigma T_i^4 + (1 - \epsilon_i) \sum F_{ij} J_j$ ; (NovaSolver Project) net  $q_i = (E_{bi} - J_i) / [(1 - \epsilon_i) / (\epsilon_i A_i)]$ .
- View factor rules: reciprocity  $A_i F_{ij} = A_j F_{ji}$ ; (arxiv) summation  $\sum F_{ij} = 1$ . (NovaSolver Project)
- Small object in large enclosure:  $q = \epsilon_1 A_1 \sigma (T_1^4 - T_{surr}^4)$ .

**Interview Q (space, quantitative):** "Size a radiator to reject 1 kW at 300 K to deep space (4 K)." **Answer outline:**  $q = \epsilon \sigma A (T_{rad}^4 - T_{space}^4)$ . With  $\epsilon \approx 0.85$ ,  $\sigma = 5.67 \times 10^{-8}$ ,  $T = 300 \text{ K}$ :  $\sigma T^4 \approx 459 \text{ W/m}^2$ .  $A \approx 1000 / (0.85 \cdot 459) \approx 2.6 \text{ m}^2$ . Note  $T_{space}^4$  is negligible. Emphasize that radiator area scales as  $1/T^4$  — running hotter dramatically shrinks the radiator, a key spacecraft design lever. Also mention absorbed environmental loads — solar (total solar irradiance =  $1361 \text{ W/m}^2$ , per Kopp & Lean 2011; TSIS-1 on ISS measures  $1361.6 \pm 0.3 \text{ W/m}^2$ ), albedo, and planetary IR — reduce net rejection.

**Interview Q (gotcha):** "Why does multilayer insulation (MLI) work in space but not in air?" **Answer:** MLI's many low-emissivity reflective shields attenuate *radiative* transfer (each layer reflects most incident radiation). It only works in vacuum because it does nothing to stop conduction/convection — in air, gas conduction and convection short-circuit the shields. Effective emittance drops with number of layers; solid conduction through spacers is the parasitic loss to minimize.

## 12. Boiling & Condensation, Critical Heat Flux — Fusion Critical

**Concepts.** Pool boiling curve (Nukiyama): natural convection → onset of nucleate boiling (ONB) → nucleate boiling (ScienceDirect) (very high  $h$ ) → **critical heat flux (CHF / DNB / boiling crisis)** → transition boiling → Leidenfrost minimum → film boiling.

### Governing/results.

- **CHF (Zuber / Kutateladze hydrodynamic-instability model):**  $q''_{max} = C \cdot h_{fg} \cdot \rho_v^{1/2} [\sigma g (\rho_l - \rho_v)]^{1/4}$ . Zuber's hydrodynamic-instability derivation gives the constant  $C = \pi/24 \approx 0.131$  (equal to the Kutateladze number), the intermediate value Zuber recommended (Mudawar et al., Purdue, *Int. J. Heat Mass Transfer* 2018); a value up to  $\sim 0.149$  is used for large horizontal plates. CHF is a strong function of pressure via  $\rho_v$ ,  $\sigma$ ,  $h_{fg}$ . (Nuclear Power)
- Nucleate boiling: Rohsenow correlation. Film boiling: Berenson (Advanceseng) (adds radiation across vapor film).
- CHF is the design-limiting event: beyond it, a vapor blanket forms,  $h$  collapses, wall temperature spikes → burnout.

**Interview Q (fusion):** "A tokamak divertor sees  $\sim 10 \text{ MW/m}^2$ . Why is CHF the governing constraint, and how do you push it higher?" **Answer:** If local wall heat flux exceeds CHF in

the water coolant channels, the surface transitions to film boiling, h crashes, and the tungsten/copper components melt (burnout). Per the ITER Organization, the divertor vertical targets are rated for 10 MW/m<sup>2</sup> steady state and 20 MW/m<sup>2</sup> for slow transients (the 20 MW/m<sup>2</sup> capability is specified for slow transients up to ~10 s for hundreds of events). You raise CHF via high subcooling, high mass flux/velocity (~10 m/s), elevated pressure (PWR-like), swirl/twisted tapes, and geometry like **hypervapotrons** (JET demonstrated ~15–30 MW/m<sup>2</sup> removal with water <11 m/s at ~1 MPa) or screw tubes for DEMO (~10 MW/m<sup>2</sup>). Helium-cooled W/SiC monoblocks are an alternative (~5–10 MW/m<sup>2</sup>).

**Interview Q (space gotcha):** "How does boiling change in microgravity?" **Answer:** Without buoyancy, vapor bubbles don't detach and rise; they coalesce into a large bubble that blankets the surface, (nasa) degrading heat transfer and lowering the effective CHF. Bubble departure relies on other mechanisms (Marangoni/thermocapillary convection, (ScienceDirect) forced flow). This is why spacecraft two-phase thermal systems require forced-flow (pumped) loops and careful vapor management rather than pool boiling.

### 13. Heat Exchangers

**Concepts.** Two design/analysis methods: LMTD (when all four terminal temperatures known → sizing) and  $\epsilon$ -NTU (when outlet temps unknown → performance).

#### Governing/results.

- Overall:  $q = UA\Delta T_{lm}$ ;  $\Delta T_{lm} = (\Delta T_1 - \Delta T_2) / \ln(\Delta T_1 / \Delta T_2)$ ; (Mechsimulator) apply correction factor F for cross/multi-pass.
- Counterflow always achieves higher effectiveness than parallel flow for the same NTU (driving force stays higher along the length). (Simulations4all)
- $\epsilon$ -NTU:  $NTU = UA/C_{min}$ ; (Mechsimulator)  $C_r = C_{min}/C_{max}$ ;  $\epsilon = q/q_{max}$ ,  $q_{max} = C_{min}(T_{h,i} - T_{c,i})$ . (Ansys Innovation Sp...) Effectiveness relations  $\epsilon = f(NTU, C_r)$  per configuration.
- $1/(UA) = \text{sum of convective} + \text{wall} + \text{fouling resistances}$ .

**Interview Q:** "When do you use  $\epsilon$ -NTU instead of LMTD?" **Answer:** Use  $\epsilon$ -NTU when outlet temperatures are unknown (Ansys Innovation Sp...) (rating an existing HX) — LMTD would require iteration. (Ansys Innovation Sp...) Use LMTD when all terminal temps are known and you're sizing (finding required UA/area). (Heat Transfer Today) They're mathematically equivalent (boundary-value vs initial-value framings of the same ODEs). (Heat Transfer Today)

### 14. Conjugate Heat Transfer

**Concepts.** Simultaneous solution of conduction in a solid coupled to convection in adjacent fluid, with continuity of temperature and heat flux at the interface (no assumed h). Essential where wall conduction and fluid convection are comparable — e.g., turbine blade cooling, fusion first-wall/monoblock, electronics.

**Interview Q:** "When must you do a conjugate analysis rather than applying a convection coefficient  $h$ ?" **Answer:** When the solid's thermal resistance is comparable to the fluid's (so wall temperature is non-uniform and you can't prescribe an isothermal or iso-flux boundary), when lateral conduction in the wall redistributes heat, or when  $h$  itself depends on the wall temperature field. Fusion plasma-facing components (steep gradients through tungsten armor into a coolant) are canonical conjugate problems.

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## PART III — INDUSTRY-SPECIFIC DEEP DIVES

### 15. Fusion Energy

**Signature topics:** plasma-facing components (PFCs), divertor/first-wall high heat flux, breeding-blanket thermal-hydraulics, liquid-metal MHD, tritium breeding, neutron volumetric heating, cryogenic magnet cooling.

- **Heat loads:** First wall  $\sim 0.5\text{--}7\text{ MW/m}^2$ ; divertor  $10\text{ MW/m}^2$  steady,  $20\text{ MW/m}^2$  slow transient (ITER). Neutron heating is *volumetric* (deposited through material thickness), unlike surface plasma flux — a key distinction interviewers probe.
- **Breeding blankets & coolants:** Concepts include WCLL (water-cooled Li-Pb), DCLL (dual-coolant Li-Pb, He + PbLi), HCPB (helium-cooled pebble bed), and all-liquid FLiBe (e.g., ARC). Coolant choices: helium (inert, no MHD, but poor  $h \rightarrow$  needs high pressure/velocity), water (excellent  $h$ , PWR-like, but activation/safety), liquid metals (great  $h$  + tritium breeding but severe MHD).
- **Tritium breeding ratio (TBR):** must exceed  $\sim 1.05\text{--}1.15$  for self-sufficiency; achieved via  ${}^6\text{Li}(n,\alpha)\text{T}$ . (Taylor & Francis Online) PbLi designs use  ${}^6\text{Li}$  enrichment (e.g., up to 90%); (arxiv) FLiBe needs less enrichment ( $\sim 20\%$  saturates TBR). (arxiv) FLiBe ( $2\text{LiF} \cdot \text{BeF}_2$ ) doubles as breeder, coolant, moderator, and neutron shield. (ResearchGate)
- **PbLi properties (interview-ready):** at  $\sim 500^\circ\text{C}$ ,  $\sigma \approx 0.77 \times 10^6\text{ S/m}$ ,  $\rho \approx 9300\text{ kg/m}^3$ ,  $\mu \approx 1.15 \times 10^{-3}\text{ Pa}\cdot\text{s}$ ,  $k \approx 15\text{--}16\text{ W/m}\cdot\text{K}$ .
- **Cryogenic magnets:**  $\text{Nb}_3\text{Sn}/\text{NbTi}$  (ScienceDirect) (or REBCO HTS) superconducting coils. Per the ITER Organization, the magnets become superconducting when cooled with supercritical helium in the range of  $4\text{ K}$  ( $-269^\circ\text{C}$ ), supplied at  $\sim 0.5\text{--}0.6\text{ MPa}$ ; three cryogenic cold boxes each rated  $25\text{ kW}$  at  $4\text{ K}$  serve the plant, and the magnet system stores  $51\text{ GJ}$  of magnetic energy. Thermal shields are cooled by forced He at  $\sim 80\text{ K}$ . Heat leaks (radiation + conduction + nuclear) must be intercepted; quench protection dumps stored magnetic energy to external resistors. (arxiv)

**Interview Q (integrative):** "Compare helium vs liquid-metal cooling for a blanket." **Answer:** He: chemically inert, no MHD interaction, transparent to tritium permeation issues differently, but low density/heat capacity  $\rightarrow$  needs  $\sim 8\text{ MPa}$  and high velocity, large pumping power, poor volumetric heat removal. Liquid metal (PbLi/Li): excellent heat transfer, breeds tritium in situ, high-temperature capability, but MHD pressure drop  $\propto B^2$  is severe (needs insulating FCIs), plus corrosion and tritium-recovery challenges. DCLL tries to get the best of both.

## 16. Materials Science / Engineering

**Signature topics:** thermal conductivity/diffusivity, thermal stress, thermal shock & fatigue, radiation effects, high-T behavior, processing.

- **Thermal stress:** constrained thermal expansion  $\sigma = E\alpha\Delta T/(1-\nu)$  (biaxial). CTE mismatch at interfaces (e.g., W armor on Cu heat sink) drives cyclic stress and fatigue.
- **Thermal shock resistance parameter:**  $R = \sigma_f k(1-\nu)/(E\alpha)$  (arxiv) — high strength & conductivity, low modulus & expansion resist shock. A second parameter  $R'$  (nih) includes  $k$  explicitly for high heat-transfer (high-Bi) conditions. Materials with low  $k$  and high  $\alpha$  (many ceramics) are shock-vulnerable. (ScienceDirect)
- **Biot-number dependence:** merit index for thermal-shock survival differs between high-Bi (sudden surface quench, strength-limited) and low-Bi (gradual, flux-limited) regimes. (ScienceDirect)
- **Radiation effects:** neutron irradiation reduces thermal conductivity (defect/phonon scattering), causes swelling, embrittlement, transmutation — degrading thermal performance over life. Critical for fusion structural materials (Eurofer97 RAFM steel, W, SiC/SiC, V-4Cr-4Ti).

**Interview Q (materials gotcha):** "Rank thermal shock resistance: window glass, tungsten, alumina, copper." **Answer:** Use  $R = \sigma_f k(1-\nu)/(E\alpha)$ . Copper and tungsten (high  $k$ , moderate strength) resist shock far better than glass and alumina (low  $k$ , brittle, high  $E$ ) despite ceramics' high-T capability. Glass fails readily under quench because low  $k \rightarrow$  steep gradients  $\rightarrow$  high transient tensile surface stress on a brittle solid. Emphasize the *ratio*, not any single property.

## 17. Space / Aerospace

**Signature topics:** vacuum radiation thermal control, MLI, heat pipes, radiators, re-entry aerothermal heating, ablation, cryogenic propellant management, microgravity effects.

- **Vacuum reality:** no convection; heat balance is absorbed environmental (solar  $1361 \text{ W/m}^2$ , albedo, planetary IR) + internal dissipation = radiated  $\sigma\epsilon T^4$ . Passive control: coatings (tune  $\alpha_{\text{solar}}/\epsilon_{\text{IR}}$  ratio), MLI, radiators; active: heat pipes, pumped loops, louvers.
- **Heat pipes:** two-phase capillary devices; evaporator wicks liquid, vapor travels to condenser, capillary action returns liquid — nearly isothermal, no pumping power. Effective conductance far exceeds solid metal.
- **Re-entry aerothermal heating:** stagnation-point convective heating from **Fay-Riddell** (1958):  $q_s \propto (\rho_e \mu_e)^{0.4} (dU_e/dx)^{0.5} (h_0 - h_w)$ , with Lewis-number/dissociation terms; heating scales  $\sim$  with  $1/\sqrt{\text{nose radius}}$   $\rightarrow$  blunt bodies reduce peak heating (Allen-Eggers blunt-body principle). ITER divertor flux ( $10\text{--}20 \text{ MW/m}^2$ ) is roughly ten times that of a spacecraft re-entering Earth's atmosphere (per ITER Organization).

- **Ablative TPS:** sacrificial material pyrolyzes/sublimes, carrying away energy and injecting gas into the boundary layer (transpiration/blockage) — used on capsules; reusable systems use insulating tiles / active cooling.
- **Cryogenic propellant management:** minimize boil-off via MLI, vapor-cooled shields, low-conductance supports; microgravity complicates phase separation and ullage.

**Interview Q (space gotcha):** "Why are re-entry vehicles blunt, not pointed?" **Answer:** A blunt body creates a strong detached bow shock that stands off the surface, dumping most of the kinetic energy into heating the shock-layer gas (radiated/convected away) rather than into the vehicle. Stagnation heating scales as  $\sim 1/\sqrt{R_{\text{nose}}}$ , so a large nose radius *reduces* peak heat flux. A sharp nose would concentrate extreme heating and burn through. This is the Allen-Eggers insight.

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## PART IV — CONCEPTUAL "GOTCHA" QUESTIONS & HOW TO REASON

1. **Biot vs Nusselt:** Both are  $hL/k$ , but Biot uses the *solid's*  $k$  (internal vs surface resistance in a body); Nusselt uses the *fluid's*  $k$  (convective vs conductive transport in the fluid). Never interchange them.
  2. **Conductivity vs diffusivity:**  $k$  sets the *steady-state* gradient;  $\alpha = k/\rho c_p$  sets the *transient* speed. "Which cools faster?"  $\rightarrow$  compare  $\alpha$ .
  3. **Emissivity = absorptivity?** Only at the same wavelength/temperature (Kirchhoff's law). Solar absorptivity (short-wave)  $\neq$  IR emissivity (long-wave) — this is exactly how spacecraft coatings are engineered (low  $\alpha_s$ /high  $\epsilon$ ).
  4. **Choked flow:** Lowering back pressure below choking doesn't increase mass flow.
  5. **MHD walls:** Conducting walls  $\rightarrow \Delta p \propto B^2$ ; insulating  $\rightarrow \Delta p \propto B$ . If you forget the wall-conductance ratio, you'll be off by orders of magnitude.
  6. **Turbulent correlation on laminar flow:** Dittus-Boelter needs  $Re > 10^4$ ; laminar  $Nu$  is constant.
  7. **Natural convection in space:** doesn't exist without gravity — a trap for candidates reciting terrestrial correlations.
  8. **Radiation linearization:** For small  $\Delta T$  you can linearize  $\sigma T^4$  to a radiative  $h_r = 4\epsilon\sigma T^3$ , but for large  $\Delta T$  (re-entry, furnaces) the full  $T^4$  is essential.
  9. **Pr of liquid metals  $\ll 1$  vs oils  $\gg 1$ :** changes which boundary layer is thicker and which correlations apply.
  10. **Fins can hurt:** if  $Bi$  based on fin is  $> 1$  (adding a fin where  $h$  is already high relative to fin  $k$ ), a fin can *increase* resistance. Fins help only when the added surface area outweighs the added conduction resistance.
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## PART V — HOW TO APPROACH OPEN-ENDED / ESTIMATION QUESTIONS

**A repeatable framework:**

1. **State the goal & unknown** and list what governs it (which transport mode dominates?).
2. **Identify the regime with dimensionless numbers.** Re (laminar/turbulent?), Ra vs forced (which convection?), Bi (lumped or distributed?), Ma (compressible?), Ha/N (is MHD braking dominant?), Pr (which BL is thicker?).
3. **Write the governing balance** (energy balance, momentum balance) and simplify by dropping negligible terms — justify each.
4. **Non-dimensionalize / scale** to expose the controlling group.
5. **Plug order-of-magnitude numbers;** carry units; sanity-check against limiting cases ( $B \rightarrow 0$ ,  $Bi \rightarrow 0$ , etc.).
6. **State assumptions and their validity range explicitly** — interviewers reward this more than a precise number.

**Worked estimation (fusion):** "Roughly, what pumping penalty does a strong field impose on a liquid-metal blanket channel?"  $\rightarrow$  Recognize MHD dominates ( $N = \sigma B^2 L / \rho U \gg 1$ ). Write  $\Delta p = k \sigma u B^2 L$  with conducting-wall  $k = c / (1 + c)$ . Plug PbLi numbers ( $\sigma \sim 0.8 \times 10^6$ ,  $u \sim 0.1$ ,  $B \sim 4\text{--}10$  T,  $L \sim 0.1$  m): get  $\sim 10^5\text{--}10^6$  Pa/m. Conclude: prohibitive without insulating inserts; motivates FCIs. This demonstrates regime identification + governing equation + estimate + design conclusion.

**Worked estimation (space):** "How big a radiator for a 5 kW satellite payload?"  $\rightarrow$  Only radiation rejects heat.  $A = Q / (\epsilon \sigma T^4)$ . At 300 K,  $\sim 1.3$  m<sup>2</sup>/kW  $\rightarrow \sim 6\text{--}7$  m<sup>2</sup> (plus margin for absorbed environment). Conclude: either large deployable radiators or run electronics hotter ( $\text{area} \propto 1/T^4$ ).

## Recommendations (How to Study)

1. **Week 1 — Fundamentals:** Re-derive continuity, Navier–Stokes term-by-term, the heat equation, and the boundary-layer equations. Memorize the dimensionless-number table with *physical meanings*, not just formulas. Benchmark: can you explain any group in one sentence?
2. **Week 2 — Correlations & methods:** Drill Dittus–Boelter/Sieder–Tate/Gnielinski (and their validity limits), the pool-boiling curve + Zuber CHF, LMTD vs  $\epsilon$ -NTU, radiation networks + view-factor algebra. Benchmark: solve 10 quantitative problems without notes.
3. **Week 3 — Your target sector:** For **fusion**, master MHD (Hartmann/Shercliff layers,  $c/(1+c)$  vs  $3\pi/8Ha$ ,  $N$  and  $Ha$ ), divertor CHF, blanket coolant trade-offs, PbLi/FLiBe properties. For **space**, master radiation sizing, MLI physics, Fay–Riddell/blunt-body reasoning, microgravity boiling. For **materials**, master transient conduction, thermal-shock merit indices,  $k$ -vs- $\alpha$ , irradiation degradation.
4. **Week 4 — Integration & estimation:** Practice the open-ended framework aloud. Do mixed "gotcha" drills. Benchmark: you can identify the dominant dimensionless group and set up any problem in  $<60$  seconds.

5. **Sources to study from:** Incropera & Bergman (heat transfer), Cengel (both), White / Fox & McDonald / Kundu (fluid mechanics), Bejan (convection), Anderson (compressible & hypersonic), plus Müller & Bühler *Magnetofluidynamics in Channels and Containers* (MHD) and Davidson (MHD) for fusion, and Gilmore/SMAD or the NASA TFAWS materials for spacecraft thermal control.

**Thresholds that change your prep:** If the role emphasizes CFD, add turbulence-model depth (wall treatment,  $y^+$ , model selection) and hands-on solver experience. If it's experimental, add instrumentation/uncertainty. If it's design, emphasize the estimation framework and safety margins over derivations.

## Caveats

- Correlation constants and property values vary by source and temperature; always cite the correlation and its validity range in an interview rather than a bare number. PbLi electrical conductivity in particular is quoted anywhere from  $0.72\text{--}0.80 \times 10^6$  S/m at  $\sim 500^\circ\text{C}$  depending on the fit; the Schulz/ORNL correlations and the Mas de les Valls et al. (*J. Nucl. Mater.* 376, 2008) database are the standard references. (Note: the Schulz thermal-conductivity fit is nominally valid only to  $\sim 625$  K, so  $k \approx 15\text{--}16$  W/m·K at 773 K is a short extrapolation.)
- The MHD  $c/(1+c)$  result is the fully-developed, thin-wall, high-Ha limit; real blanket manifolds have complex 3-D effects (expansions, bends, fringing fields) that require CFD and can dominate the total drop.
- Heat-flux capability numbers (hypervapotron  $15\text{--}30$  MW/m<sup>2</sup>, monoblock  $5\text{--}10$  MW/m<sup>2</sup>) are design/test values under specific conditions, not universal guarantees; the ITER  $10/20$  MW/m<sup>2</sup> divertor figures are the official steady-state/slow-transient design ratings.
- This guide targets breadth for interview readiness; for actual design work consult the primary textbooks and validated codes.